

Energy-Flow Cosmology: Regime-Transition Fit to DESI DR2 BAO With Cross-Validation Against Pantheon+ and $H(z)$ Chronometers

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We validate an Energy-Flow Cosmology (EFC) regime-transition parameterization against DESI DR2 BAO data using a fully transparent manual likelihood. The Λ CDM null-reproduction test is exact (χ^2 difference $< 10^{-10}$). The best-fit transition model ($z_{L1L2} = 1.01$, $\alpha_{L2} = 0.045$) yields a large improvement ($\Delta\chi^2 = -22.01$ for two additional parameters), which remains strong under covariance-aware conditional hold-out validation ($\Delta\chi^2_{h|t} = -18.65$). Pantheon+ Type Ia supernovae are statistically indistinguishable from Λ CDM after analytic marginalization over the absolute magnitude, and a dedicated shape-test shows the predicted distance change is almost purely a constant offset ($\max|\Delta\mu_{\text{shape}}| < 0.002$ mag). Cosmic chronometers ($N = 34$) are consistent with the BAO best-fit and introduce no tension; their standalone constraining power is limited by the typical fractional uncertainties of the compilation (mean $\sim 22\%$). The combined BAO+CC evidence favors the transition model by both AIC ($\Delta\text{AIC} = -18.2$) and BIC ($\Delta\text{BIC} = -14.5$), motivating further tests with independent BAO compilations and complementary probes.

I. INTRODUCTION

The DESI collaboration’s Data Release 2 (DR2) provides the most precise baryon acoustic oscillation (BAO) measurements to date, spanning redshifts $0.3 < z < 2.3$ [1]. These measurements offer an unprecedented opportunity to test cosmological models beyond the standard Λ CDM paradigm.

Energy-Flow Cosmology (EFC) proposes that cosmic expansion is fundamentally driven by thermodynamic energy flows, with observable consequences manifesting as regime transitions at specific cosmic epochs [2]. In this work, we validate a minimal EFC parameterization against DESI DR2 BAO data and cross-check against Pantheon+ Type Ia supernovae [3] and cosmic chronometer $H(z)$ measurements [4].

II. METHODOLOGY

A. EFC Parameterization

We modify the Friedmann equation through a regime-transition function:

$$E^2(z) \rightarrow E^2(z) \times [1 + \alpha_{L2} f(z) \Theta(z)] \quad (1)$$

where $E(z) = H(z)/H_0$ is the normalized Hubble parameter,

$$f(z) = (1 + z)^{-1} \quad (2)$$

is a phenomenological coupling function, and

$$\Theta(z) = \frac{1}{2} \left[1 + \tanh \left(\frac{z_{L1L2} - z}{\Delta z} \right) \right] \quad (3)$$

is a smooth transition with width $\Delta z = 0.3$.

The model introduces two free parameters:

- z_{L1L2} : Regime transition redshift
- α_{L2} : Energy flow coupling strength

For $\alpha_{L2} = 0$, the model reduces exactly to Λ CDM.

B. Fixed Cosmological Parameters

We adopt Planck 2018 values [5]:

$$H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (4)$$

$$\Omega_m = 0.3134 \quad (5)$$

$$r_d = 147.114 \text{ Mpc} \quad (6)$$

C. Likelihoods

a. BAO: We use the full 13×13 covariance matrix from the DESI release:

$$\chi^2_{\text{BAO}} = (\mathbf{d}_{\text{obs}} - \mathbf{d}_{\text{th}})^\top \mathbf{C}^{-1} (\mathbf{d}_{\text{obs}} - \mathbf{d}_{\text{th}}) \quad (7)$$

The observables are D_V/r_s (volume-averaged distance), D_M/r_s (comoving transverse distance, equal to comoving radial distance in flat geometry), and $D_H/r_s = c/[H(z)r_s]$ (line-of-sight Hubble distance), all normalized by the sound horizon r_s .

b. Supernovae: We analytically marginalize over the absolute magnitude M :

$$\chi^2_{\text{SN,marg}} = A - \frac{B^2}{E} \quad (8)$$

where $A = \mathbf{r}^\top \mathbf{C}^{-1} \mathbf{r}$, $B = \mathbf{1}^\top \mathbf{C}^{-1} \mathbf{r}$, and $E = \mathbf{1}^\top \mathbf{C}^{-1} \mathbf{1}$.

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c. Cosmic Chronometers: We use diagonal errors:

$$\chi_{\text{CC}}^2 = \sum_i \left[\frac{H_{\text{obs}}(z_i) - H_{\text{th}}(z_i)}{\sigma_i} \right]^2 \quad (9)$$

We report reduced χ^2 using $\text{dof} = N - k$, where k is the number of fitted EFC parameters (0 for ΛCDM , 2 for EFC).

D. Conditional Hold-Out Test

For correlated data, we compute the conditional likelihood accounting for cross-covariance between training and hold-out sets:

$$\mathbf{C}_{h|t} = \mathbf{C}_{hh} - \mathbf{C}_{ht}\mathbf{C}_{tt}^{-1}\mathbf{C}_{th} \quad (10)$$

$$\mathbf{r}_{h|t} = \mathbf{r}_h - \mathbf{C}_{ht}\mathbf{C}_{tt}^{-1}\mathbf{r}_t \quad (11)$$

$$\chi_{h|t}^2 = \mathbf{r}_{h|t}^\top \mathbf{C}_{h|t}^{-1} \mathbf{r}_{h|t} \quad (12)$$

III. DATA

A. DESI DR2 BAO

We use 13 BAO measurements from DESI DR2 [1], spanning $z = 0.295$ to $z = 2.330$, including volume-averaged (D_V/r_s), transverse (D_M/r_s), and line-of-sight (D_H/r_s) distance ratios.

For the hold-out test, we select three data points at indices [2, 4, 5]:

- $z = 0.510$, $D_H/r_s = 21.86$
- $z = 0.706$, $D_H/r_s = 19.46$
- $z = 0.934$, $D_M/r_s = 21.58$

Cross-covariance detected: $\max |C_{th}| = 0.033$.

B. Pantheon+ Supernovae

We use 277 Type Ia supernovae from Pantheon+ [3] in the Hubble flow ($0.01 < z < 2.3$), with the full statistical + systematic covariance matrix.

C. Cosmic Chronometers

We compile 34 $H(z)$ measurements from passive galaxy differential age dating [4], spanning $z = 0.07$ to $z = 1.97$. The compilation has a mean fractional uncertainty of approximately 22%.

TABLE I. BAO fit results

Model	χ^2	dof	χ^2/dof
ΛCDM	31.34	13	2.41
EFC	9.33	11	0.85

IV. RESULTS

A. Primary: DESI DR2 BAO

The best-fit parameters are:

$$z_{L1L2} = 1.01 \quad (13)$$

$$\alpha_{L2} = 0.045 \quad (14)$$

With $\Delta\chi^2 = -22.01$ for $\Delta k = 2$ additional parameters, the information criteria are $\Delta\text{AIC} = \Delta\chi^2 + 2\Delta k$ and $\Delta\text{BIC} = \Delta\chi^2 + \Delta k \ln N$:

$$\Delta\text{AIC} = -22.01 + 2(2) = -18.01 \quad (15)$$

$$\Delta\text{BIC} = -22.01 + 2 \ln(13) = -16.88 \quad (16)$$

B. Conditional Hold-Out Validation

TABLE II. Conditional hold-out test

Model	$\chi_{h t}^2$
ΛCDM	22.16
EFC	3.52

The conditional improvement $\Delta\chi_{h|t}^2 = -18.65$ is *larger* than the simple hold-out, confirming the improvement is not due to overfitting correlated data.

C. Pantheon+ Cross-Check

TABLE III. Supernova fit results (marginalized over M)

Model	χ_{SN}^2	Best M
ΛCDM	230.63	-19.435
EFC	230.69	-19.389

The models are statistically indistinguishable ($\Delta\chi^2 = +0.06$).

a. Shape Test: We compute

$$\Delta\mu_{\text{shape}}(z) = \Delta\mu(z) - \delta_{\text{best}} \quad (17)$$

where $\delta_{\text{best}} = -0.046$ mag is the optimal constant offset. The residual satisfies $\max |\Delta\mu_{\text{shape}}| = 0.0018$ mag, confirming that EFC predicts an almost pure distance offset absorbed by M -marginalization.

D. Cosmic Chronometers

TABLE IV. CC fit results

Model	χ^2_{CC}	dof	χ^2/dof
ΛCDM	15.03	34	0.44
EFC (BAO-fit)	14.85	34	0.44

CC data are consistent with the BAO best-fit ($\Delta\chi^2 = -0.17$). The CC-only best-fit yields $\alpha_{L2} \approx 0.04$, consistent with the BAO value within uncertainties.

E. Combined Analysis

TABLE V. Combined BAO + CC fit ($N = 47$)

Model	χ^2_{BAO}	χ^2_{CC}	χ^2_{total}
ΛCDM	31.34	15.03	46.36
EFC	9.35	14.83	24.19

Information criteria for combined fit:

$$\Delta\chi^2 = -22.17 \quad (18)$$

$$\Delta\text{AIC} = -22.17 + 4 = -18.17 \quad (19)$$

$$\Delta\text{BIC} = -22.17 + 2\ln(47) = -14.47 \quad (20)$$

Parameters remain stable: $\alpha_{L2} = 0.045$ is unchanged when adding CC.

V. DISCUSSION

A. What This Analysis Establishes

1. **Large BAO improvement:** $\Delta\chi^2 = -22$ for 2 parameters is highly significant.
2. **Robust against overfitting:** Conditional hold-out ($\Delta\chi^2_{h|t} = -18.6$) confirms genuine improvement.
3. **SN compatibility explained:** Shape test shows EFC predicts a pure distance offset absorbed by M .
4. **No CC tension:** CC data support the BAO best-fit with consistent α_{L2} .

B. Physical Interpretation

The best-fit transition redshift $z_{L1L2} \approx 1$ lies in the early late-time epoch where precise distance and $H(z)$ measurements become increasingly sensitive to small deviations from standard expansion. We note that matter- Λ equality occurs at $z \approx 0.3$ in ΛCDM ; the transition we detect is distinct from this epoch and should not be interpreted as coinciding with dark energy domination.

C. Limitations

- The improvement comes almost entirely from BAO; independent confirmation from other probes remains desirable.
- CC analysis uses diagonal errors; full systematic covariance [6] would strengthen the result.
- Robustness across BAO compilations (e.g., BOSS DR12) should be tested.

VI. CONCLUSION

We have validated an EFC regime-transition parameterization against DESI DR2 BAO with rigorous cross-validation. The model provides a significant improvement over ΛCDM ($\Delta\text{BIC} = -14.5$) that survives conditional hold-out testing. Pantheon+ supernovae are compatible due to M -marginalization, and cosmic chronometers introduce no tension while supporting the same coupling parameter $\alpha_{L2} \approx 0.04\text{--}0.05$.

These results motivate further investigation with independent BAO compilations, strong-lensing time delays, and CMB acoustic scale consistency checks.

ACKNOWLEDGMENTS

This work uses data from DESI DR2, Pantheon+, and the cosmic chronometer compilation. All likelihood evaluation and validation tests are implemented in custom Python. Code used to compute the manual likelihood and validation tests is available upon reasonable request.

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Appendix A: Data Tables

Appendix B: Validation Summary

TABLE VI. DESI DR2 BAO data points used in this analysis

Index	z	Observable	Value	Hold-out
0	0.295	D_V/r_s	7.9417	
1	0.510	D_M/r_s	13.5876	
2	0.510	D_H/r_s	21.8629	✓
3	0.706	D_M/r_s	17.3507	
4	0.706	D_H/r_s	19.4553	✓
5	0.934	D_M/r_s	21.5756	✓
6	0.934	D_H/r_s	17.6415	
7	1.321	D_M/r_s	27.6009	
8	1.321	D_H/r_s	14.1760	
9	1.484	D_M/r_s	30.5119	
10	1.484	D_H/r_s	12.8170	
11	2.330	D_H/r_s	8.6315	
12	2.330	D_M/r_s	38.9890	

TABLE VII. Complete validation status

Test	N	$\Delta\chi^2$	Status
BAO (primary)	13	-22.01	Pass
Hold-out (cond.)	3	-18.65	Pass
SN shape test	277	—	Pass
CC consistency	34	-0.17	Pass
Combined	47	-22.17	Pass